Set Properties

→ 1 to 1 Correspondence

Two Sets P and Q
Every element of P can be Uniquely
Paired with element of Q.

- $P = \{a, b\}$ $Q = \{c, d\}$

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- Set
  - Finite
  - Infinite
    - Countable
    - Uncountable

\# Finite Set : There is 1 to 1 Corr.
between Set elements and Some other
Set K ($K \in N$)

$S = \{a, b, c, d, e\}$ ← Finite Set

→ $K = \{5, 6, 7, 8, 9\}$

→ Countable Infinite : There is 1 to 1 Corr. between a Set and Set of Natural Numbers (N)

$A = \{x \mid x \text{ is } +ve \text{ even No.}\}$

$= \{2, 4, 6, 8, 10 \ldots\ldots\}$

$N = \{1, 2, 3, 4 \ldots\ldots\}$

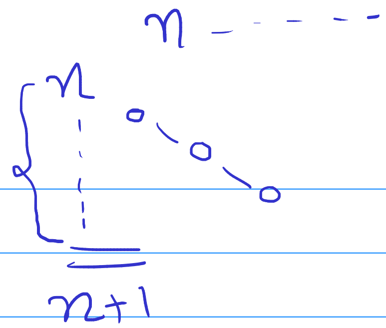
→ Uncountable Infinite

There is no 1 to 1 mapping between a Set and N.

→ Ex: S = Set of all Real No. between 0 and 1.

n →

	C	V	S
A	y	No	y
B	No	No	y
C	y	y	y
D	No	y	No



~ D is different than A, B, C.

~> Diagonal Argument

"Certain object is not one of the given objects using the fact that this object is diff. from each of the given object in at least one way"

Ex Proove that ^set of all Real Nos between 0 and 1 is Uncountable Infinite.

-> Proof by Contradiction.

-> We assume that the Set is Countably Infinite.

~> We list all Nos between 0 and 1

$\to 0.\ a_{11}\ a_{12}\ a_{13}\ a_{14} \ldots$ n

$\to 0.\ a_{21}\ a_{22}\ a_{23} \ldots$

$\to 0.\ a_{31}\ a_{32}\ a_{33}\ a_{34} \ldots$

$\vdots$

n

n+1

$0.\ b_1 b_2 b_3 \ldots$

* Properties Related to Set Cardinality

1. $|P \cup Q| \le |P| + |Q|$
2. $|P \cap Q| \le \min(|P|, |Q|)$
3. $|P \oplus Q| = |P| + |Q| - 2 \cdot |P \cap Q|$
4. $|P - Q| \ge |P| - |Q|$

* Principle of Inclusion & Exclusion

- $|A1 \cup A2| = |A1| + |A2| - |A1 \cap A2|$

- $|A1 \cup A2 \cup A3| = |A1| + |A2| + |A3| - |A1 \cap A2|$
  $- |A2 \cap A3| - |A3 \cap A1| + |A1 \cap A2 \cap A3|$

* **Principle of Mathematical Induction.**

* For a given Statement involving Natural number n. if we can show that

Basis ~ Statement is True for $n = n_0$

Induction step ~ Statement is true for $n = k+1$ assuming Statement is true for $n = k$ $(k \geq n_0)$

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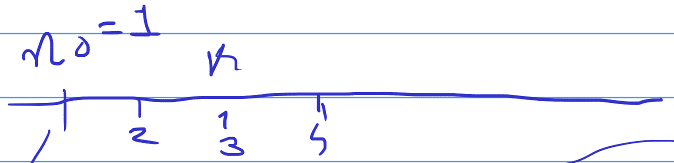

Proove that Statement is true for n_0

Assumed Statement is True for $k = 3$

Proove – Statement is True for $k+1$ (4)

Problem:

Coins $\to$ 3, 5

Any amount of 8 or above can be made using only coins of 3 and 5.

$n_0 = 8 = 3 + 5$

$9 = 3 + 3 + 3$

$10 = 5 + 5$

$11 = 5 + 3 + 3$

$12 = 3 + 3 + 3 + 3$